

Tetris row packing

Tetris-lite assigns each cell to its nearest row, then left-packs like overlap removal

The idea

- Tetris is simpler than Abacus but moves cells farther on this toy
- Contrast disp six versus Abacus disp four
- Trade simpler control flow against a tighter displacement budget and slightly higher HPWL

Pseudocode

- Tetris pseudocode is shorter than Abacus on purpose
- First assign each cell to its nearest row and freeze that choice
- Then reuse the per-row left pack from overlap removal
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Same teaching seed, same numbers as overlap removal: displacement six, HPWL thirty-two
- Write the compare line in the sketch so you remember Abacus spends search to cut

Algorithm sketch — try these

```
INPUT: positions, widths, rows Y[], fixed macros
OUTPUT: legal packing (shelf / Tetris-lite)
for each movable c: y ← nearest row (freeze)
then per-row left pack (see overlap removal)
GOLDEN: A@4 B@6 C@8 on y=2; disp=6; HPWL=32
COMPARE: Abacus disp=4 (more search)
```

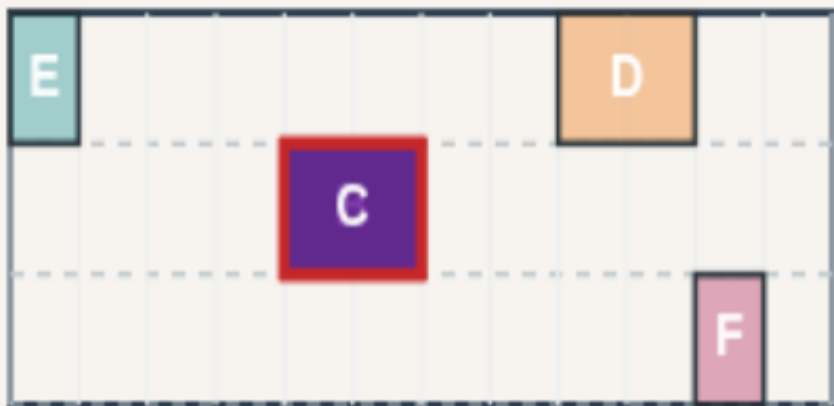
Same overlap seed

TETRIS ROW PACK

Same overlap seed

Step 1 / 5 · overlap-seed · module02-07-tetris-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Tetris-lite assigns each cell to its nearest row, then left-packs within the row—same engine as overlap removal. Start from the triple stack at (4, 2).

- Nearest row from float y
- Per-row left pack
- No cross-row trial

seed: OVERLAP
legal: false

Same overlap seed

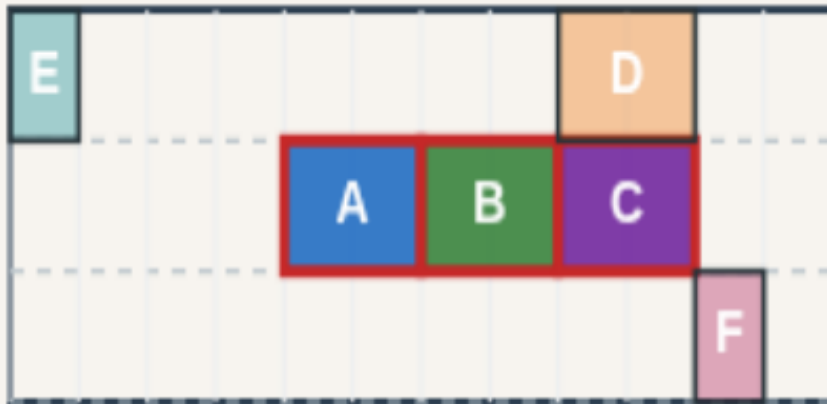
Nearest row, then left pack

TETRIS ROW PACK

Nearest row, then left pack

Step 2 / 5 · nearest-row · module02-07-tetris-row-pack

chip 12×6 · 3 rows · site 1



Explanation

A, B, and C stay on middle row two after snap. Sort by x and pack: A at four, B at six, C at eight—identical to overlap removal on this instance.

- Row locked after snap
- Shelf pack left-to-right
- Simpler control flow

all on y=2
pack order A,B,C

Nearest row, then left pack

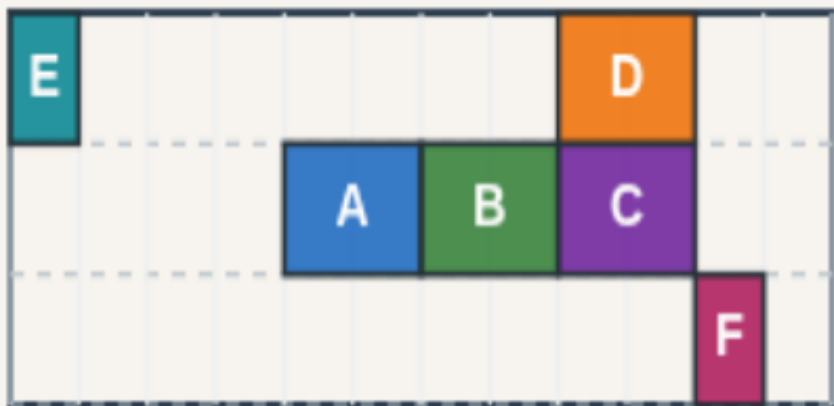
Result: disp 6, HPWL 32

TETRIS ROW PACK

Result: disp 6, HPWL 32

Step 3 / 5 · tetris-result · module02-07-tetris-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Tetris legalizes with displacement six and HPWL thirty-two—the overlap-removal golden. Same coordinates, same metrics: this is the global legalize path in later labs.

- legal: true
- disp: 6
- HPWL: 32

disp: 6
HPWL: 32
measured: 6

Result: disp 6, HPWL 32

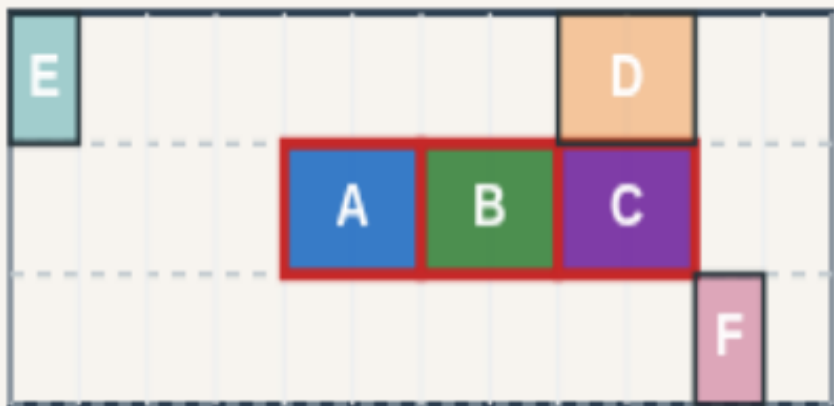
Contrast Abacus disp 4

TETRIS ROW PACK

Contrast Abacus disp 4

Step 4 / 5 · contrast-abacus · module02-07-tetris-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Abacus spreads A, B, C across three rows for displacement four. Tetris keeps them on one row and moves farther in x—six total L1 units.

- Abacus: cross-row, disp 4
- Tetris: single row, disp 6
- HPWL favors Tetris here

```
abacusDisp: 4  
tetrisDisp: 6  
Δ disp: 2
```

Contrast Abacus disp 4

Tradeoff: simpler vs better displacement

TETRIS ROW PACK

Tradeoff: simpler vs better displacement

Step 5 / 5 · tradeoff · module02-07-tetris-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Tetris is easier to implement and slightly better on HPWL for this toy. Abacus is the choice when displacement budget is tight—preview the detailed-vs-global lab.

- Tetris = global legalize lite
- Abacus = detailed legalize lite
- Pick by disp vs HPWL budget

tetrisHpwl: 32

abacusHpwl: 38

Tradeoff: simpler vs better displacement

Browser lab track

- In the browser lab track, open the **tetris-row-pack** lab from the tools shelf
- Open the interactive lab
- Reveal golden is study-only
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse `tiny\_legal.json`, run the algorithm with deterministic coordinates
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module